

ALAN GOMEZ

Universidad del Valle de Guatemala (UVG), Guatemala City

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📍 Zone 16, Guatemala

in LinkedIn

🐙 Github



EXPERIENCE

Assistant - D-Hive

University of the Valley of Guatemala

📅 Jan 2024 – Nov 2024 📍 Guatemala City, Guatemala

- Conducted training sessions and safety workshops for university students.
- Performed maintenance on 3D printers, ensuring optimal operation and minimizing downtime.
- Designed components using laser cutters for various projects.

Team Leader - Parasitic Extraction Project

University of the Valley of Guatemala

📅 Sep 2025 – Nov 2025 📍 Guatemala City, Guatemala

- Led a team to successfully complete a 3-bit multiplier in IC design.
- Characterized the multiplier and analyzed its performance, ensuring accuracy and efficiency.
- Coordinated tasks and documentation for project completion within a 2-month timeline.

Designer Builder - Sumo Robot

University of the Valley of Guatemala

📅 Mar 2023 – Aug 2023 📍 Guatemala City, Guatemala

- Designed, programmed, and assembled a competitive sumo robot.
- Won the university competition, defeating 18 participants through strategic design and programming.
- Managed all aspects of the project over 6 months, from concept to competition-ready robot.

TECHNICAL SKILLS

- **CAD / Design:** Autodesk Inventor, FreeCAD, Laser Cutting, 3D Printing
- **Programming Languages:** C, C++, Python, Microcontroller Programming, Verilog
- **Software / Tools:** MATLAB, Git/GitHub, LaTeX, Microsoft Excel, Linux
- **Cloud / Networking:** Basic AWS, Datacom concepts
- **IC Design:** Fundamentals of IC Design, Multiplier Design, Characterization

HOBBIES

- Studying Object-Oriented Programming (OOP) and Cloud Computing.

PERSONAL SKILLS

- Independent and self-motivated, with a proactive approach to problem-solving.
- Initiative-driven, capable of identifying and implementing solutions efficiently.
- Intermediate English proficiency: strong reading skills, particularly with technical documentation; moderate listening comprehension; able to engage in casual conversations with developing fluency.

EDUCATION

Bachelor in Electronics Engineering - 85% Current GPA

University of the Valley of Guatemala

📅 Jan 2022 – May 2027

PROJECTS

IoT Servo Control with Adafruit

- Implemented IoT system to control 4 servos using an ATmega328P microcontroller via Adafruit dashboard.
- Programmed in C, integrating hardware and cloud-based control for remote operation.
- Project duration: 2 months, delivered November 2023.

32-bit MIPS Processor on FPGA

- Designed and implemented a 32-bit MIPS processor on Nanoboard 3000 using Verilog.
- Performed simulation and deployment on FPGA, validating functionality and performance.
- Project duration: 2 months, delivered November 2025.

Campus Network Design and Implementation (Huawei Routers/Switches)

- Designed and configured network topology for a fictitious university campus using Huawei routers and switches.
- Implemented the configuration physically on hardware, testing connectivity and network performance.
- Developed skills in routing, switching, and network deployment in a real hardware environment.